VEMU INSTITUTE OF TECHNOLOGY

AUTOMATIC WINDOW OPENING BASED ON TEMPERATURE

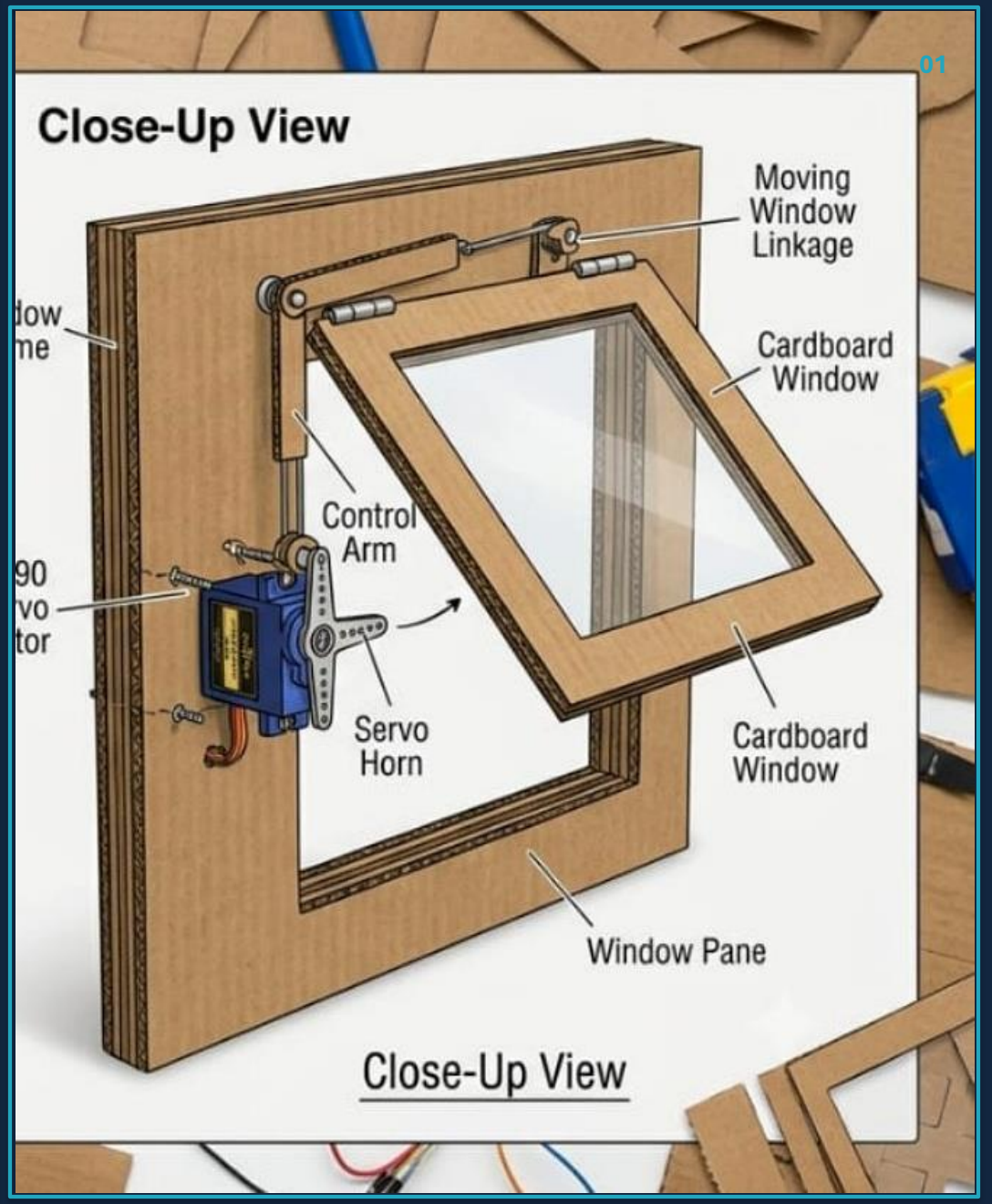
Arduino UNO • DHT11 • SG90 Servo

GUIDE

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TEAM MEMBERS

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Components Used

Hardware required to sense temperature and automate the window mechanism



01 Arduino UNO

Main controller that reads the sensor and controls the se

02 DHT11 Sensor

Measures the surrounding temperature and sends the re

03 SG90 Servo Motor

Rotates to move the window linkage between open and

04 Breadboard

Provides a convenient platform for connecting the sensor an

05 Jumper Wires

Carry power and signal connections between Arduino ar

06 USB Cable

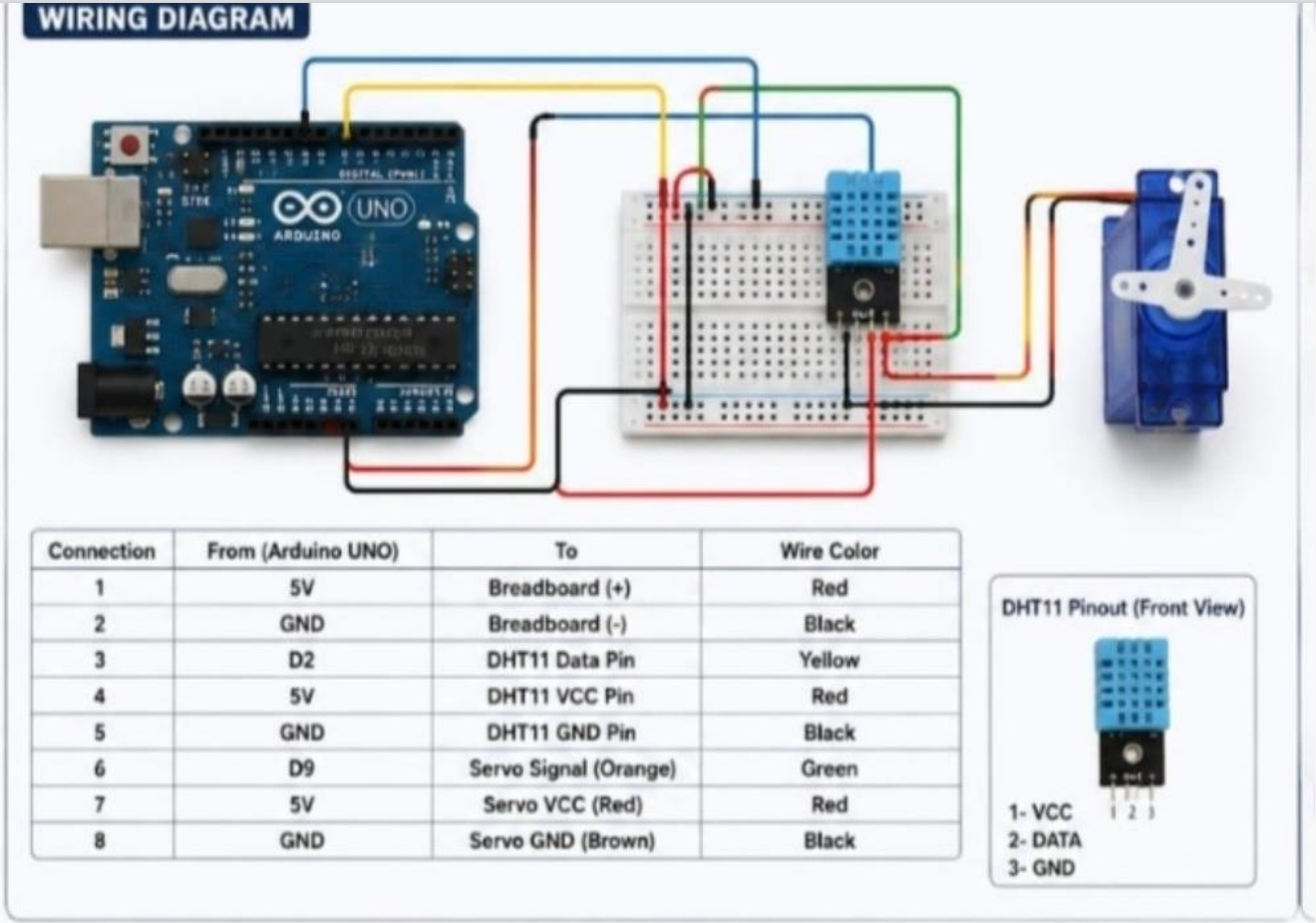
Supplies power and allows the Arduino program to be up

07 Window Model

Physical prototype on which the automatic opening mechanism is demonstrated.

Circuit Connection

Sensor input and servo output are connected to the Arduino UNO



Connection	From (Arduino UNO)	To	Wire Color
1	5V	Breadboard (+)	Red
2	GND	Breadboard (-)	Black
3	D2	DHT11 Data Pin	Yellow
4	5V	DHT11 VCC Pin	Red
5	GND	DHT11 GND Pin	Black
6	D9	Servo Signal (Orange)	Green
7	5V	Servo VCC (Red)	Red
8	GND	Servo GND (Brown)	Black

KEY CONNECTIONS

5V	→	DHT11 VCC	Power
GND	→	DHT11 GND	Ground
D2	→	DHT11 DATA	Temperature signal
D9	→	Servo SIGNAL	Control signal
5V	→	Servo VCC	Power
GND	→	Servo GND	Ground

Signal flow

DHT11 → Arduino → SG90

Working of the Project

A simple closed-loop process: sense → decide → move

1

SENSE

DHT11 continuously reads the surrounding temperature.

→

2

COMPARE

Arduino compares the reading with the preset threshold.

→

3

OPEN

If temperature is above 30°C, the servo rotates to open the window.

→

4

CLOSE

When temperature falls to 30°C or below, the servo returns and closes it.

TEMPERATURE RULE	Above 30°C → OPEN WINDOW	30°C or below → CLOSE WINDOW



FINAL PROTOTYPE

Smart Window Opening System

Our cardboard house model demonstrates how temperature sensing

AUTOMATIC

Window movement is controlled by the servo motor.

TEMPERATURE-BASED

DHT11 provides the input needed for the decision.

LOW-COST PROTOTYPE

Arduino and simple components make the idea easy to demonstrate.

A practical step toward smarter and more responsive homes.